USN											

"24CS350"

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Computer Science and Engineering

III Semester: CIE-3/Test-2

Discrete Mathematical Structures

Duration:1 Hr 15 min. Date:09.01.2026 Max. Marks:30

NOTE : PART- A is Compulsory, PART- B has internal choice

PART-A

Q.NO	CO	CD	QUESTION	MARKS
1	2	L3	Let A = {1,2,3} B={w,x,y,z} C = {4,5,6} Where R1⊆ AxB , R2 ⊆ BxC and R3 ⊆ BxC defined as R1 = {(1, w) (3, w) (2, x) (1, y)} R2= {(w, 5) (x, 6) (y, 4) (y, 6)} R3 = {(w, 4) (w, 5) (y, 5)} verify which of the following is TRUE a) R1 ∘ (R2∪R3) = (R1 ∘ R2) ∪ (R1 ∘ R3) b) R1 ∘ (R2∩R3) = (R1 ∘ R2) ∩ (R1 ∘ R3) c) M (R1 ∘ R2) = M(R1). M(R2) d) M(R1).M(R2)!= M(R2). M(R1)	08
2	4	L2	Explain the following with suitable example. a) Complement graph b) Subgraph c) Euler graph d) Hamiltonian graph e) Trail and path	07

PART-B

3	2	L3	Let R be a relation defined on A= {1,2,3,4,5,6,7,8,9,10,11,12} by $(x,y)\epsilon R$ if and only if $x-y$ is multiple of 5. a) Verify whether R is an equivalence relation. b) If yes, Find the partition of A induced by R.	07
4	4	L2	Define Isomorphism in graphs. Verify whether the graphs in Figure-1 are Isomorphic. If yes, map their edges and vertices.	08
			OR	
5	2	L3	Draw the Hasse diagram for the divisibility relation on a set A = {2,3,5,6,7,11,12,35,385} and find i) maximal and minimal element ii) greatest and least element iii) LUB {7,11} and LUB {5,7} iv) GLB {2,12} and GLB {2,3}	07
6	4	L2	Define Bipartite and Complete Bipartite Graphs with suitable example. Explain Kuratowski's graphs.	08

P.T.O

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Use the knowledge needed to test the logic of the program.	L3
CO2	Demonstrate the concept of relations and be able to determine their properties.	L3
CO3	Explain and apply decode group codes based on the various methods.	L3
CO4	Identify different types of graphs such as Isomorphism, Bipartite, planar, Hamiltonian paths and cycles, Euler trails and circuits based on their properties.	L3
CO5	Explain different types of trees and apply tree traversal techniques like preorder and postorder.	L3

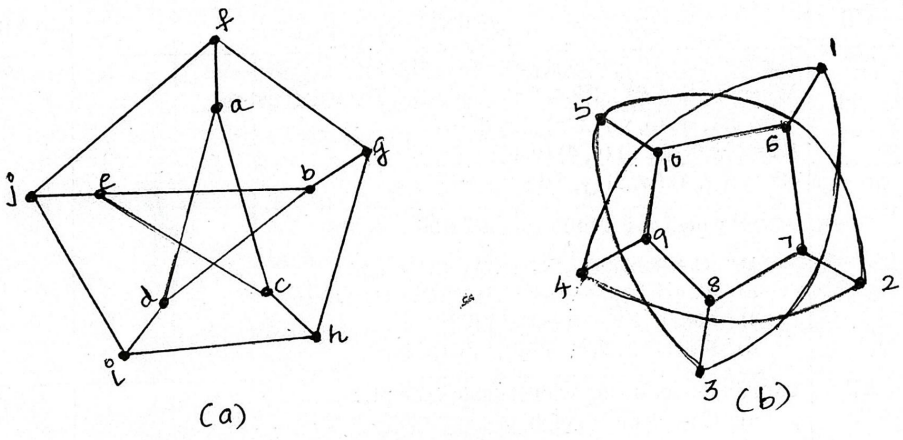


Figure-1.